

MACHINE LEARNING FOR QUANTUM ERROR CORRECTION

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Description:

The advancements of the field of quantum computing and its promises of exponential speedups for challenging problems are being hindered by quantum errors caused by decoherence (quantum properties affected by the environment) and gate errors. Quantum error corrections (QEC) are techniques to solve this problem and maintain the coherence of quantum computations. In this project, we plan to apply deep learning techniques to provide a less computation expensive yet effective way of overcoming these issues. Specifically, we will use supervised learning algorithms for predicting and correcting quantum state errors using machine learning analysis of noisy measurement data from quantum circuits simulated in Google's Cirq environment.

Data Summary:

Since there's no bench mark data, we will generate a dataset made from scratch using the help of Google's Cirq environment. Cirq is Google's opensource python software package for writing, manipulating, and optimizing quantum circuits. Basically, we will be simulating quantum circuits with and without noise (bit-flips, phase-flips etc).

Input labels will be noisy measurement outcomes from quantum circuits affected by bit-flip, phase-flip, and depolarizing noise. Probability of noise can be set between (0 % - 20 %).

Output labels will have corresponding clean measurement outcomes from the same circuits with zero noise. The data will be presented in a table of binary numbers describing the qubit states. We will have sufficient data variability to train a stable model and study issues such as class imbalances and potential correlations between types of errors.

Methods:

We will use a deep learning model to learn error patterns and predict correct quantum states. The proposed method (could be changed during large scale practical implementation) includes

1. Dataset generation: Using Cirq to simulate quantum circuits with controlled noise injections.
2. Feature Engineering: Extracting relevant patterns from measurement outcomes.
3. Neural Network Model: Training a deep learning classifier (example, a feedforward neural network or a convolutional neural network) to map noisy states to corrected states.
4. Evaluation Metrics: Assessing performance using accuracy, F1-score, and error reduction compared to traditional QEC methods.

Preliminary results:

We have managed to simulate a simple two-qubit Bell state circuit using Cirq and inserted bit-flip noise at a 10% chance. Bell states are the four states that can be created when two qubits are maximally entangled and the way we can achieve this is using one CNOT and Hadamard quantum gates. This circuit is the simplest form of showing quantum entanglement. Initial results show that our deep learning model trained on a small dataset can predict the original uncorrupted quantum state with an accuracy of about 76%. Further optimizations and larger, diverse dataset could improve these results.

References:

1. Google Cirq Documentation: <https://quantumai.google/cirq>
2. Benchmarking Machine Learning Models for Quantum Error Correction [Link](#)